

CTA L Ridership Analysis Data Sources, Cleaning, and Preparation

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Data sources

CTA L daily ridership

Data provided by the CTA accessed through the Chicago Data Portal contains daily totals of L station entries spanning from August 2016 to August 2024. The raw set contains columns station ID, station name, date, day type, and rides.

- **station ID** unique identifier of L station
- **station name** name of the L station
- **date** date in format mm/dd/yyyy
- **day type** W = weekday, A = Saturday, U = Sunday/holiday
- **rides** number of rides

CTA L Stations geographic data

Additional data provided by the CTA includes coordinates of CTA L stations and its station ID, which are the only two columns used for this project. Also includes station name, train routes, address, and accessibility.

- **the_geom** geographic location of the L station using ‘WGS84’ coordinate system
- **STATION ID** unique identifier of L station

U.S. Census Bureau Census tract geographic data

Data provided by the U.S. Census Bureau contains polygon data, mapping borders of Chicago’s census tract areas in 2000. Although the borders of Chicago’s census tracts have changed over the last two decades to accommodate changing population densities in certain areas, they have remained generally stable, and have not drastically changed. This project will use the polygon data, census tract identifier, and corresponding community area (neighborhood) identifier.

- **the_geom** polygon data, which maps borders of census tract, also using ‘WGS84’ coordinate system
- **CENSUS_TRA** census tract unique identifier
- **TRACT_COMM** community area unique identifier

American Community Survey (ACS) 5-year data by census tract

Data recorded by the ACS in 2023 contains income data and other demographic information of Americans living in each Chicago community area/neighborhood. Only income buckets columns and community name will be used.

- **Community Area** name of community area
- **Under \$25,000** count of community members with household income under 25000
- **\$25,000 to \$49,999** count of community members with household income between 25000 and 49999
- **\$50,000 to \$74,999** count of community members with household income between 50000 and 74999
- **\$75,000 to \$125,000** count of community members with household income between 75000 and 125000
- **\$125,000 +** count of community members with household income over 125000

Data cleaning and preparation

For the CTA L daily ridership raw dataset (“CTA-Ridership-L-Station-Entries_daily-totals.xlsx”), I reformatted the station id unique identifier to match station id columns in other data sets. The clean set is stored as “daily_totals.xlsx”.

- $VALUE(RIGHT([station_id],4))$ formula used in station_id_clean to reformat station_id. Takes last four digits of the original station id and VALUE() changes data type to numeric, removing any leading zeroes.

CTA L station geographic data from “CTA-L-Stations.csv” was stored as “station_info.xlsx” after being exported as an excel workbook.

U.S. Census Bureau Census tract geographic data from “Census_Tracts.csv” was also exported as an excel workbook and saved to “census_tracts.xlsx”. Empty and broken data records were identified and removed using Power Query.

- Used built in PQ filters and error detection to remove null rows.

A table of neighborhood codes were also used to cross reference neighborhood code/community number and community name.

American Community Survey (ACS) 5-year data by census tract data set from “ACS_5_Year_Data_by_Community_Area.c” is saved as “census_data.xlsx”. Added column community number by lookup formula.

“stations_tracts_joined.xlsx” is a dataframe exported from R script “geospatial_data_cleaning.R”, containing joined data from station_info and census_tracts.

1. Converted “the_geom” points in station_info dataframe to spatial data using the sf package. Wkt = 1 indicates to use the first column of the dataframe. Crs = 4326 tells the function to use the world Geodetic System 1984, the global geographic coordinate system.
2. Converted “the_geom” multipolygon points in census_tracts dataframe to spatial data. When plotted with stations_points, geographic positions of each CTA L station relative to census tract borders can be observed.
3. Using st_join, with x argument being stations_points, y argument being cen_tracts_points_poly, and join = st_within, I obtain a joined dataframe. st_within tells the function to align each station point to the census tract it is inside of, and left join the corresponding data from cen_tracts_points_poly to each CTA station in station_points.

4. `na_stations` was created to list stations that did not fall into a Chicago census tract. 26 stations are stored and removed from the main dataframe.
5. The dataframe of type sf “`stations_tracts_joined_points`” is changed back to a standard dataframe using `st_drop_geometry()`.

To compile the “master” worksheet, I used first generated connections to all previously cleaned sets in Excel’s built in Power Query.

- In PQ, inner join with `daily_totals` and `stations_tracts_joined` excludes daily totals that were recorded in stations outside of Chicago tracts.
- Still in PQ, used left outer joins to append other sheets and select relevant columns.